

1.

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

Your Code

```
1 #include <stdio.h>
2 void swap(int *a, int *b) {
3     int temp = *a;
4     *a = *b;
5     *b = temp;
6 }
7 int partition(int arr[], int low, int high) {
8     int pivot = arr[high];
9     int i = low - 1;
10    for (int j = low; j < high; j++) {
11        if (arr[j] < pivot) {
12            i++;
13            swap(&arr[i], &arr[j]);
14        }
15    }
16    swap(&arr[i + 1], &arr[high]);
17    return i + 1;
```

Input

stdin input

Output

Sorted array: 9 18 23 32 50 61

2.

Write a program that implements the bubble sort

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[100], n = 0;
4     while (scanf("%d", &a[n]) == 1) {
5         n++;
6     }
7     for (int i = 0; i < n; i++) {
8         for (int j = 0; j < n - i - 1; j++) {
9             if (a[j] > a[j + 1]) {
10                int temp = a[j];
11                a[j] = a[j + 1];
12                a[j + 1] = temp;
13            }
14        }
15    }
16    for (int i = 0; i < n; i++) {
17        printf("%d", a[i]);
```

Input

23 11 9 10 5 4

Output

4 5 9 10 11 23

3.

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[] = {9, 5, 1, 4, 3};
4     int n = 5;
5     for (int i = 1; i < n; i++) {
6         int key = a[i];
7         int j = i - 1;
8
9         while (j >= 0 && a[j] > key) {
10             a[j + 1] = a[j];
11             j--;
12         }
13         a[j + 1] = key;
14     }
15     for (int i = 0; i < n; i++) {
16         printf("%d", a[i]);
17         if (i < n - 1) {
```

Input

stdin input

Output

1 3 4 5 9

4.

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

Your Code

```
1 #include <stdio.h>
2 void merge(int arr[], int l, int m, int r) {
3     int i, j, k;
4     int n1 = m - l + 1;
5     int n2 = r - m;
6     int L[n1], R[n2];
7     for (i = 0; i < n1; i++)
8         L[i] = arr[l + i];
9     for (j = 0; j < n2; j++)
10         R[j] = arr[m + 1 + j];
11     i = 0;
12     j = 0;
13     k = l;
14     while (i < n1 && j < n2) {
15         if (L[i] <= R[j]) {
16             arr[k] = L[i];
17             i++;
```

Input

stdin input

Output

3 9 10 27 38 43 82